TSM 098 | INDEPENDENT PRACTICE

Linear equations

While TSM 099 studies Topic 27. Review: Topic 24, Topic 25, Topic 26.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: two-term and short linear equations

1. Solve and check in the original equation.

$$5x + 7 = 37$$

2. Solve and check in the original equation.

$$-3x + 1 = 7$$

3. Solve and check in the original equation.

$$-6x - 6 = 0$$

4. Solve and check in the original equation.

$$3x + 2 = -1$$

5. Solve and check in the original equation.

$$-3x + 4 = 13$$

6. Solve and check in the original equation.

$$-4x - 1 = -9$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: variable terms on both sides

7. Solve and check in the original equation.

$$-5x + 2 = -2x + 5$$

8. Solve and check in the original equation.

$$4x + 3 = -2x + 6$$

9. Solve and check in the original equation.

$$2x + 6 = -2x + 7$$

10. Solve and check in the original equation.

$$2x + 2 = x + 2$$

11. Solve and check in the original equation.

$$-3x - 6 = x + 4$$

12. Solve and check in the original equation.

$$2x + 5 = -2x + 8$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: equations with parentheses

13. Solve and check in the original equation.

$$-3(x+3)-4=2x-8$$

14. Solve and check in the original equation.

$$-3(x-4)+6=27$$

15. Solve and check in the original equation.

$$2(3x+2)-2=-3x+20$$

16. Solve and check in the original equation.

$$3(2x-4)-4=-x-23$$

17. Solve and check in the original equation.

$$-2(2x+1)-6=x-23$$

18. Solve and check in the original equation.

$$-2(x-1)+5=3x-3$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: mixed equations and special cases

19. Solve and check in the original equation.

$$5x - 3 = 2x + 8$$

20. Solve and check in the original equation.

$$-4x + 9 = 3x - 2$$

21. Solve and check in the original equation.

$$2(3x - 4) - 3 = 2x + 5$$

22. Solve and check in the original equation.

$$-3(2x + 1) + 2 = 3x - 7$$

23. Solve and check in the original equation.

$$2(x + 3) = 2x + 6$$

24. Solve and check in the original equation.

$$3(x - 2) = 3x + 1$$

TSM 098 | ANSWER KEY

Linear equations

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: two-term and short linear equations

1. Solve and check in the original equation.

$$5x + 7 = 37$$

$x = 6$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$-3x + 1 = 7$$

$x = -2$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-6x - 6 = 0$$

$x = -1$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$3x + 2 = -1$$

$x = -1$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$-3x + 4 = 13$$

$x = -3$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$-4x - 1 = -9$$

$x = 2$

Check by substituting into the original equation.

TSM 098 | ANSWER KEY

Linear equations

While TSM 099 studies Topic 27. Review: Topic 24, Topic 25, Topic 26.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: variable terms on both sides

7. Solve and check in the original equation.

$$-5x + 2 = -2x + 5$$

$x = -1$

Check by substituting into the original equation.

8. Solve and check in the original equation.

$$4x + 3 = -2x + 6$$

$x = \frac{1}{2}$

Check by substituting into the original equation.

9. Solve and check in the original equation.

$$2x + 6 = -2x + 7$$

$x = \frac{1}{4}$

Check by substituting into the original equation.

10. Solve and check in the original equation.

$$2x + 2 = x + 2$$

$x = 0$

Check by substituting into the original equation.

11. Solve and check in the original equation.

$$-3x - 6 = x + 4$$

$x = -\frac{5}{2}$

Check by substituting into the original equation.

12. Solve and check in the original equation.

$$2x + 5 = -2x + 8$$

$x = \frac{3}{4}$

Check by substituting into the original equation.

TSM 098 | ANSWER KEY

Linear equations

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: equations with parentheses

13. Solve and check in the original equation.

$-3x - 13 = 2x - 8$

$x = -1$

Check by substituting into the original equation.

14. Solve and check in the original equation.

$-3x + 18 = 27$

$x = -3$

Check by substituting into the original equation.

15. Solve and check in the original equation.

$6x + 2 = -3x + 20$

$x = 2$

Check by substituting into the original equation.

16. Solve and check in the original equation.

$6x - 16 = -x - 23$

$x = -1$

Check by substituting into the original equation.

17. Solve and check in the original equation.

$-4x - 8 = x - 23$

$x = 3$

Check by substituting into the original equation.

18. Solve and check in the original equation.

$-2x + 7 = 3x - 3$

$x = 2$

Check by substituting into the original equation.

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Linear equations

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: mixed equations and special cases

19. Solve and check in the original equation.

$$5x - 3 = 2x + 8$$

$$x = \frac{11}{3}$$

Check by substituting into the original equation.

20. Solve and check in the original equation.

$$-4x + 9 = 3x - 2$$

$$x = \frac{11}{7}$$

Check by substituting into the original equation.

21. Solve and check in the original equation.

$$6x - 11 = 2x + 5$$

$$x = 4$$

Check by substituting into the original equation.

22. Solve and check in the original equation.

$$-6x - 1 = 3x - 7$$

$$x = \frac{2}{3}$$

Check by substituting into the original equation.

23. Solve and check in the original equation.

$$2x + 6 = 2x + 6$$

All real numbers (the two sides are identical).

24. Solve and check in the original equation.

$$3x - 6 = 3x + 1$$

No solution (the two sides cannot be equal).